

# Midterm Exam I

Due: 3 pm October 20, 2021

Name \_\_\_\_\_

Student ID \_\_\_\_\_

## VERSION A

### Instructions:

- NO interactions and communications with anyone in any form—**academic integrity and honesty** is important
- Answer all questions using a blue/black pen on A4 papers
- Upload your answers and the first page of the exam to WM5 by **3:10 pm** (10 points off if uploading after 3:20 pm, and submission after 3:20 pm will not be accepted)
- Read and sign the statement below

By signing this, I declare that the submitted exam was produced independently by me, without help from others.

Signature \_\_\_\_\_

Microeconomics **Midterm 1**, Fall 2021

**Name and Student ID:**

Part One (40 points):

1. (5 points) What does the sign of the cross-price elasticity of demand between two goods tell us about the nature of the relationship between those goods?

財貨 X 財貨 Y 的交叉彈性可表為  $\epsilon_{xy}$

when  $\epsilon_{xy} > 0 \Rightarrow$  兩者為替代品

$= 0 \Rightarrow$  中性財

$< 0 \Rightarrow$  互補品

2. (5 points) A 10 percent increase in the price of automobiles reduces the quantity of automobiles demanded by 8 percent. What is the price elasticity of demand for automobiles?

$$\epsilon_A = \frac{\Delta Q/Q}{\Delta P/P} = \frac{-8\%}{10\%} = -\frac{4}{5} = -0.8 \#$$

3. (5 points) In class, we talk about Raj Chetty's research on tax salience. Using a field experiment in a grocery store, he finds that consumers underreact to taxes that are not salient—posting tax-inclusive price tags reduces demand by 8 percent. What is the implication of his result on the relationship between legal incidence and economic incidence of sales tax?

In economic incidence, 課稅在消費者 or 廠商的結果是相同的

In legal incidence, 當課稅在消費者上時, 反而會減少他們的消費量

$\Rightarrow$  legal incidence 與 economic incidence 會有衝突

4. (10 points) Demand for product A is given by  $Q^D = 50 - 0.5P$ . What is the price elasticity of demand when the price is \$60? Is demand inelastic or elastic? If a seller of product A wants to increase revenue, should she increase or decrease the price of product A?

$$Q^D = 50 - 0.5 \times 60 = 20$$

$$\epsilon = \frac{\partial Q}{\partial P} \times \frac{P}{Q} = -0.5 \times \frac{60}{20} = -\frac{3}{2} \Rightarrow \text{elastic}$$

seller needs to lower the price of product if he wants to increase revenue.

5. (5 points) An economist estimates that the elasticity of teenage employment to minimum wage is -0.01. Explain what it means in one sentence. Suppose the economist also estimates that the elasticity of teenage wages to minimum wage is 0.5. Can you say anything about the labor demand elasticity of teenagers?

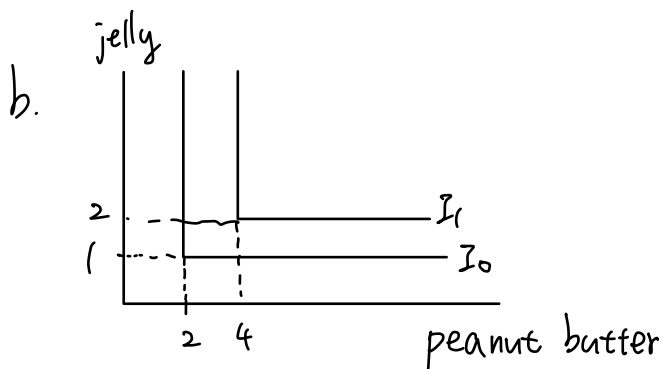
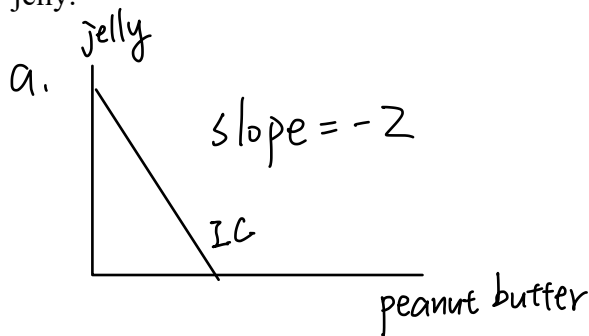
$\epsilon = -0.01$  when minimum wage increase 1%, <sup>teenage</sup> employment decrease for 0.01%

$$\epsilon = 0.5 \quad \frac{T_w}{M_w} = 0.5$$

$$\frac{T_e}{M_w} = -0.01$$

$$L_D \text{ elasticity of teenagers} = \frac{\frac{T_e}{M_w}}{\frac{T_w}{M_w}} = \frac{-0.01}{0.5} = -0.02 \#$$

6. (10 points) Draw two indifference curves for each of the following two types of consumer preferences: (a) I like both peanut butter and jelly, and always get the same additional satisfaction from an ounce of peanut butter as I do from 2 ounces of jelly. (b) I like peanut butter and jelly, but I only want 2 ounces of peanut butter for every ounce of jelly.



Part Two (80 points):

1. (25 points) Consider the market for Cheese Puff Snacks ( $Q$  is in bags of Cheese Puffs,  $P$  is in \$/bag). The demand for Cheese Puff Snacks is  $Q^D = 30 - P$  and the supply is  $Q^S = 3P - 10$ .

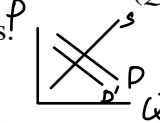
- a. (10 points) What are the equilibrium price ( $P^*$ ) and quantity ( $Q^*$ ) of cheese puffs?

$$Q^D = Q^S \Rightarrow 30 - P = 3P - 10 \Rightarrow P^* = 10 \quad Q^* = 20$$

- b. (10 points) To pay for classes about healthy snacking, the government imposes a tax on Cheese Puffs of \$4 per bag. What are the price paid by buyers ( $P_b$ ), price received by sellers ( $P_s$ ), and the number of bags of Cheese Puffs sold ( $Q_T$ ) after the tax? Calculate the tax incidence for consumers and producers!

$$Q^D = 30 - P \quad Q^S = 3P - 10$$

$$P_s = 9$$



$$P = 30 - Q^D$$

$$P_b = 9 + 4 = 13$$

$$P = 30 - Q^D - 4$$

$$Q_T = 17$$

$$P = 26 - Q^D$$

tax incidence for consumers = 3 \$ per bag  
for producers = 1 \$ per bag

$$\Rightarrow Q^D = Q^S \Rightarrow 26 - P = 3P - 10$$

$$P = 9$$

- c. (5 points) Suppose the government decides to expand the healthy snacking program and doubles the amount of the tax. Will the government double its revenue, less than double its revenue, or more than double its revenue? Why?

$$\text{Tax revenue for \$4 per bag} = 4 \times 17 = 68$$

$$\text{If tax \$8 per bag} = 8 \times 14 = 112$$

$$22 - P = 3P - 10$$

$$Q_T = 14$$

$$68 \times 2 > 112$$

$$32 = 4P$$

$\Rightarrow$  less than double

$$8 = P_s$$

its revenue

$$8 + 8 = 16 = P_b$$

b/c consumer will consume less

they will not consume  $Q_T = 17$

2. (20 points) Recessions lower incomes and reduce rental prices for beachfront properties. Suppose that the demand function for renting a beachfront property in Barcelona, Spain, during the first week of August is  $Q = 1000 - P + Y/20$ , where  $Y$  is the median annual income of the people involved in this market,  $Q$  is quantity, and  $P$  is the rental price. The supply function is  $Q = 2P - Y/20$ .

- a. (10 points) Derive the equilibrium price,  $P$ , and equilibrium quantity,  $Q$ , in terms of  $Y$ .

$$\begin{aligned}
 Q^D &= 1000 - P + \frac{Y}{20} \\
 Q^S &= 2P - \frac{Y}{20} \\
 Q^D &= Q^S \Rightarrow 1000 - P + \frac{Y}{20} = 2P - \frac{Y}{20} \\
 1000 + \frac{Y}{10} &= 3P \\
 \frac{1000}{3} + \frac{Y}{30} &= P^* \\
 Q^* &= 2\left(\frac{1000}{3} + \frac{Y}{30}\right) - \frac{Y}{20} \\
 Q^* &= \frac{2000}{3} + \frac{Y}{60} \quad \#
 \end{aligned}$$

- b. (10 points) Use calculus to derive the effect of decreased income on the equilibrium price and quantity of rental homes. [Hint: find  $dP/dY$  and  $dQ/dY$ .]

$$\text{when eqm. } P = \frac{1000}{3} + \frac{Y}{30} \quad Q = \frac{2000}{3} + \frac{Y}{60}$$

$$\frac{dP}{dY} = \frac{1}{30} \quad \text{when } Y \text{ decrease, eqm price decrease } \frac{1}{30} \% \text{ per } 1\%$$

$$\frac{dQ}{dY} = \frac{1}{60} \quad Y \quad 1\% \quad \text{quantity} \quad \frac{1}{60} \%$$

3. (35 points) Answer the following two questions on constrained maximization problems:

3.1 (20 points) Suppose Antonio gets utility from consuming two goods, burgers and fries. His utility function is given by  $U = \sqrt{BF} = B^{0.5}F^{0.5}$  where  $B$  is the number of burgers and  $F$  is the number of servings of fries he eats per day. Antonio's daily income is \$20, the price of a burger is \$5, and the price of fries is \$2.

- (10 points) Write down the constrained maximization problem for Antonio. What are Antonio's utility-maximizing quantities of burgers and fries?
- (5 points) Suppose the restaurants selling burgers and fries impose a quantity limit of one burger per day for each customer, how does this regulation affect Antonio's optimal choice? Explain.

$U = B^{0.5}F^{0.5}$   $\ln U = \frac{1}{2}(\ln B + \ln F)$   
 $V = 2 \ln U$  c. (5 points) Will your answers for the part a and b change if Antonio's utility function is given by  $V = \ln B + \ln F$ ? Explain without doing any calculations.

a.  $MRS = MRT \Rightarrow + \frac{0.5}{0.5} \times \frac{F}{B} = + \frac{5}{2} \Rightarrow 2F = 5B$   
 $5B + 2F = 20$

$$\begin{cases} 2F = 5B \\ 5B + 2F = 20 \end{cases} \Rightarrow F = 5 \quad B = 2$$

b. If only one burger, then he will spend all his money on fries

$B = 1 \quad F = 7.5$  ( $MRS > MRT$ ) he wants to consume more burger, but he can't

c. It will be the same with a and b, b/c it's a positive

monotonic transformation  $U$  and  $V$  represent the same preference

$$V = 2 \ln U$$

3.2 (15 points) Wolf's utility function is given by  $U = aq_1^{0.5} + q_2$ .

- (5 points) Show that his MRS is the same on all of his indifference curves at a given  $q_1$ .
- (10 points) For given prices and income, show how he has an interior or corner solution depends on  $a$ .

$$a. \text{ MRS} = - \frac{MU_1}{MU_2} = - \frac{\frac{1}{2} a q_1^{-0.5}}{1} = - \frac{a}{2\sqrt{q_1}}$$

at a given  $q_1$ , MRS is the same for  $-\frac{a}{2\sqrt{q_1}}$

b. when  $a$  is big, there is a interior solution

small

corner

$$\textcircled{1} a \leq 0 \Rightarrow MU_1 < 0 \quad MU_2 = 1 \\ \Rightarrow q_1 = 0 \quad q_2 = \frac{Y}{P_2}$$

$$\textcircled{2} a > 0$$

$$\begin{cases} 0.5 a q_1^{-0.5} = \frac{P_1}{P_2} \\ P_1 q_1 + P_2 q_2 = Y \end{cases}$$

$$q_1 = \left( \frac{0.5 a P_2}{P_1} \right)^2$$

$$q_2 = \frac{P_1 Y - (0.5 a P_2)^2}{P_1 P_2}$$

$$\text{if } P_1 Y - (0.5 a P_2)^2 \geq 0$$

$$\text{if } P_1 Y - (0.5 a P_2)^2 \leq 0$$

$$\Rightarrow q_1 = \frac{Y}{P_1} \quad q_2 = 0$$

by  $\textcircled{1}, \textcircled{2}$  Corner solution if  $a \leq 0$  or  $a \geq \frac{2\sqrt{P_1 Y}}{P_2}$

# Part One

1. 財貨 X 財貨 Y 的交叉彈性為  $\epsilon_{xy}$  2.  $P \uparrow 10\%$

$Q \downarrow 8\%$

當  $\epsilon_{xy} > 0 \Rightarrow$  兩者為替代<sup>需求</sup>品

$< 0$  互補品

$= 0$  相互品

$$\epsilon = \frac{-8\%}{10\%} = \frac{-4}{5} = -0.8$$

課稅在 而且他們

3. 如果消費者對於稅比較不注意 就有可能承擔較多的租稅負擔

也代表他們對於商品的需求比較沒有彈性

相反的, 廠商希望將稅部分由消費者負擔, 提高價錢的彈性比較容易達到

$\Rightarrow$  economic incidence is related to legal incidence

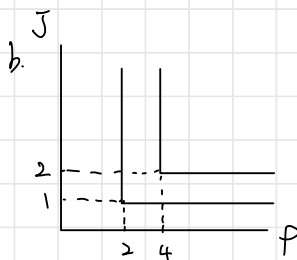
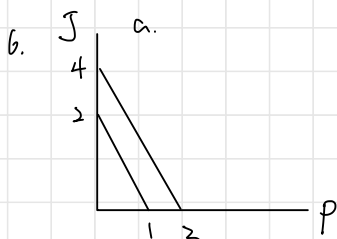
$$4. \epsilon = \frac{\partial Q}{\partial P} \times \frac{P}{Q} \Rightarrow -0.5 \times \frac{60}{20} = -\frac{3}{2}$$

She needs to decrease price of product to increase revenue.

5. When minimum wage increases 10%, teenage employment decreases 0.1%

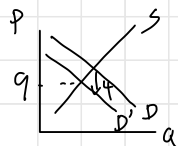
$$\epsilon = \frac{\partial \ln E^D}{\partial \ln \text{Min wage}}$$

$$L_D \text{ elasticity of teenager} = \frac{\frac{\partial \ln E^D}{\partial \ln \text{Min wage}}}{\frac{\partial \ln \text{teenage wage}}{\partial \ln \text{Min wage}}} = \frac{-0.1}{0.5} = -0.2$$





Part two



1. a  $Q^D = Q^S$

$$30 - P = 3P - 10$$

$$40 = 4P$$

$$10 = P^*$$

$$Q^* = 20$$

b.  $Q = 30 - P$

$$Q = 3P - 10$$

$$\Rightarrow P = 30 - Q$$

$$P = 30 - Q - 4$$

$$= 26 - Q$$

$$26 - P = 3P - 10$$

$$36 = 4P$$

$$Q_T = 17$$

$$9 = P \rightarrow \text{received by firms}$$

$$9 + 4 = 13 \rightarrow \text{paid by buyers}$$

$$\frac{1}{1+3} = \frac{1}{4} \rightarrow \text{tax incidence on firm}$$

$$\frac{3}{1+3} = \frac{3}{4} \Rightarrow \text{buyers}$$

c. Tax = 4 Tax Revenue =  $4 \times 17 = 68$

$$T_{GX} = 8$$

$$22 - P = 3P - 10$$

$$32 = 4P$$

$$8 = P^*$$

$$24 - 10 = 14$$

$$P_b = 16$$

$$14 \times 8 = 112$$

112 < 68 \times 2 \Rightarrow \text{b/c buyers will decrease consume, they will not consume } Q\_T = 17

2.  $Q = 1000 - P + \frac{Y}{20}$

(a)  $Q = 2P - \frac{Y}{20}$

$$1000 - P + \frac{Y}{20} = 2P - \frac{Y}{20}$$

$$1000 + \frac{Y}{10} = 3P$$

$$\frac{1000}{3} + \frac{Y}{30} = P^* \quad Q^* = \frac{2000}{3} + \frac{1}{60}Y$$

$$\frac{\frac{1}{15} - \frac{1}{20}}{\frac{20 - 15}{300}} = \frac{1}{60}$$

(b)  $\frac{dP}{dY} = \frac{1}{30}$

$$\frac{dQ}{dY} = \frac{1}{60}$$

decrease income, decrease both

P and Q

3.1 a.  $U = B^{0.5} F^{0.5}$

$$5B + 2F = 20$$

$$MRS = -\frac{MU_B}{MU_F} = -\frac{0.5F}{0.5B} = -\frac{F}{B}$$

$$MRT = -\frac{P_B}{P_F} = -\frac{5}{2}$$

$$\begin{aligned} \int MRS &= MRT \Rightarrow +\frac{F}{B} = +\frac{5}{2} \Rightarrow 2F = 5B \\ 5B + 2F &= 20 \Rightarrow 4F = 20 \\ F &= 5 \quad B = 2 \quad \# \end{aligned}$$

b. When consume only one burger  $|MRS| > |MRT|$ , he wants to consume more but he cannot. So he consume 1 burger and 7.5 fries

c.  $\ln U = 0.5 \ln B + 0.5 \ln F$

$$2 \ln U = \ln B + \ln F$$

$\Rightarrow V = 2 \ln U$  正單調變換不會影響偏好

Antonio 對漢堡薯條的偏好相同

3.2

a.  $U = a q_1^{0.5} + q_2$

$$MRS = \frac{-\frac{1}{2} a q_1^{-0.5}}{1} = -\frac{a}{2\sqrt{q_1}}$$

$\therefore$  the slope of a given  $q_1$  is independent of  $q_2$

$\therefore$  the slope at  $q_1$  must be the same on ICs

$$a. \mu_1 = \frac{1}{2} a q_1^{-0.5} \quad \text{if } a < 0 \quad \text{then } \mu_1 < 0$$

$$\mu_2 > 0$$

$$\text{Wolf will consume } q_1 = 0 \quad q_2 = \frac{Y}{P_2}$$

$$MRT = -\frac{P_1}{P_2}$$

$$\left\{ \begin{array}{l} MRT = MRS \Rightarrow + \frac{a}{2\sqrt{q_1}} = -\frac{P_1}{P_2} \\ P_1 q_1 + P_2 q_2 = Y \end{array} \right.$$

$$a P_2 = 2 P_1 \sqrt{q_1}$$

$$\left( \frac{a P_2}{2 P_1} \right)^2$$

$$q_1 = \left( \frac{0.5 a P_2}{P_1} \right)^2$$

$$q_2 = \frac{P_1 Y - (0.5 a P_2)^2}{P_1 P_2}$$

$$q_2 = \frac{Y - P_1 q_1}{P_2}$$

$$\begin{aligned} & Y - P_1 \left( \frac{0.5 a P_2^2}{P_1} \right) \\ & \quad \quad \quad P_2 \\ & = \frac{P_1 Y - 0.5 a P_2^2}{P_1 P_2} \end{aligned}$$

$$\text{if } P_1 Y \geq (0.5 a P_2)^2$$

$$\text{if } P_1 Y \leq (0.5 a P_2)^2 \Rightarrow q_1 = \frac{Y}{P_1} \quad q_2 = 0$$

$$\Rightarrow \text{corner solution } a \leq 0, \quad a \geq 2\sqrt{P_1 Y}$$